CHE-494: Biopharmaceutical Process Development and Manufacturing

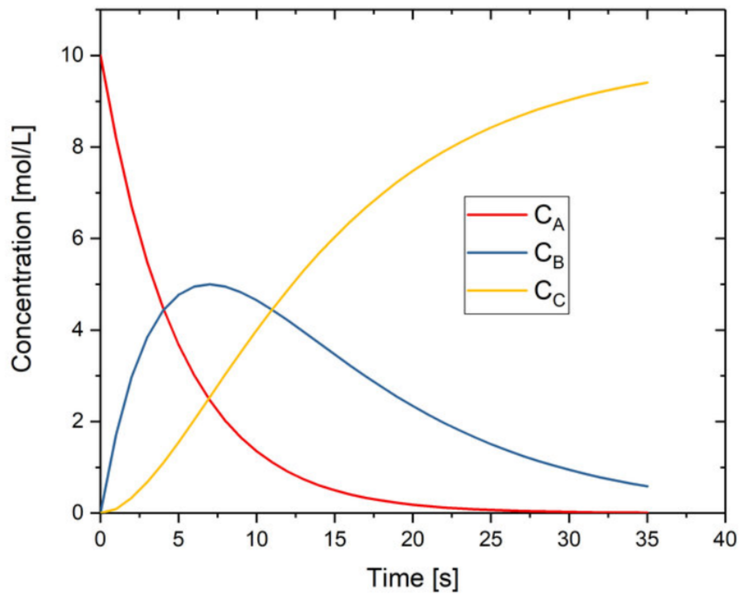
Project Assignment

Guidelines:

1. **Please do not share and discuss your assignment with others.**
2. **Two case studies italicized to tailor your answer accordingly.**
3. **Prepare a slide presentation to answer and communicate with the class.**
4. **You may please work on the questions during the course of instructions as and when pertinent classes are delivered. Hence there is no separate homework or written quizzes.**
5. **Questions cover all the course contents.**

Questions:

1. What makes you excited (or not) to work for biopharmaceutical company?
2. *A soluble small molecule drug substance (active pharmaceutical ingredient, API) has very good physical properties, namely the tight particle size distribution, great flowability and high bulk density. What drug product process would you recommend for solid dosage form? Provide a process flow chart and outline process design considerations for the unit operations involved.*
3. What do you mean by reliable process scale-up? What does it mean in the context of high shear wet granulation process?
4. What are the key considerations while developing the oral formulation for the small molecule drug substance? Why?
5. What is the most prevalent **unit process** in small molecule drug substance process? Why?
6. What are the key considerations for the design of optimal liquid-liquid extraction process? Why?
7. *Time profiling key reaction speci: reactant A, product B and the undesired product, C from a drug substance producing reaction is depicted in the following plot.*



Identify the reaction mechanism. What kind of reactor would you suggest for conducting such type of reaction with B being the product? Also, provide recommendations to carry out reliable reaction scale-up?

8. What is the most difficulty purification technique for the large molecule drug substance? Provide key considerations for the process development and reliable scale-up of the same.